BSR Transition Risk — Methodology

Reflects the code as built through P0–P3. This document specifies the flow, mathematics, and logic of the standalone transition-risk engine (`engine/transition/`) and app (`transition_app/`). It is generated to be marked up: each section states what the code does, the exact formula, the data behind it, and the judgment parameters you can challenge.

Companion to the physical-risk methodology. Screening-grade throughout — not a regulatory disclosure without specialist review.

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1. Overview

The engine decomposes the financial impact of decarbonisation on an asset or portfolio into **four channels**, each mapped to one **TCFD transition-risk category**, quantified by one **engine layer**, and routed to exactly one **financial destination** (the *non-duplication* rule):

TCFD category	Layer	Mechanism	Routes to
Policy & Legal	**L1**	(Scope 1+2) × NGFS carbon price × (1 − pass-through)	Cash-flow OpEx
Technology	**L2**	Wright's-Law cost crossover / demand collapse → logistic impairment	Cash flow + asset value
Market	**L3**	Sector shock → Leontief inverse → focal input cost	Cash-flow input cost
Reputation	**L4**	Sautner CCExposure (z-scored) × per-SD elasticities	Cash flow **or** WACC

On top of the four layers sit a **decision & assurance layer**: Monte-Carlo uncertainty, tornado sensitivity, PCAF financed emissions, Implied Temperature Rise, PACTA-style alignment, peer benchmarking, marginal-abatement-cost economics, a budget optimizer, and an IFRS S2 / ESRS E1 disclosure export.

How to run

```
pip install streamlit pandas numpy plotly openpyxl
streamlit run transition_app/Home.py            # the app
python -m pytest tests/test_transition.py -q     # the tests (all green)
```

The engine + tests need only numpy/pandas/openpyxl/pytest. `pyam` and `pymrio` are only for the data-refresh scripts (NGFS prices, EXIOBASE MRIO).

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2. Architecture & module map

```
engine/transition/
  data_loader.py        cached JSON loaders; ISO3→NGFS region; scenario→NGFS mapping
  carbon_pricing.py     L1 — carbon cost, pass-through, abatement pathway, priced fraction
  learning_curves.py    L2 — Wright's law, Lafond bands, crossover, logistic impairment
  network_propagation.py L3 — world 20×20 Leontief propagation
  network_mrio.py       L3 — 20×49 EXIOBASE MRIO + Reisch endogenous-default cascade
  cc_exposure.py        L4 — z-standardised CCExposure × per-SD elasticities + routing
  transition_engine.py  orchestrator: run_asset_transition / run_portfolio_transition
  transition_dcf.py     combined physical+transition DCF hook
  uncertainty.py        Monte-Carlo over price / pass-through / elasticity
  sensitivity.py        one-at-a-time tornado
  alignment.py          financed emissions, ITR, pathway alignment, peer benchmark
  macc.py               marginal abatement cost curves (decarbonise vs pay)
  optimizer.py          budget-constrained merit-order abatement allocation

data/transition/
  carbon_prices_ngfs.json      NGFS Phase V REMIND-MAgPIE (real, ×1.18 → USD2020)
  sector_pass_through.json     sector-median pass-through
  learning_curves.json         learning rates + LCOE base costs (IRENA 2024 / Lazard 2025)
  sector_pathways.json         demand pathways (production index, 2025=1.0)
  cc_exposure_proxy.json       sector CCExposure (real ×10³ scale) + pooled z-base + elasticitie
  io_matrix.json               20×20 world direct-requirements matrix (EXIOBASE world totals)
  io_matrix_mrio.npz + meta     20×49 EXIOBASE MRIO (980×980)
  sector_taxonomy.json         20 sectors: tech pairs, fossil flag, emission intensity
  macc.json                    per-sector abatement measures (AR6 WG3 / IEA-informed)
```

Execution flow

```
① Data Entry  →  intake per entity: sector, Scope 1/2/3, replacement value, revenue,
                 target_year, priced_pct, attribution_pct
        |
        ▼
   run_portfolio_transition(assets, scenarios, …settings…)
        |  for each (scenario, asset):
        |    L1 carbon_cost_timeline(...)          → net_carbon_opex[y]
        |    L2 compute_stranding(...)             → impairment[y], revenue_erosion[y]
        |    L3 propagate (world | mrio+cascade)   → indirect_input_cost[y]
        |    L4 compute_exposure_premium(...)      → revenue_modifier[y] OR wacc_bps
        |    total_cf[y] = L1 + L2_rev + L3 + L4 ;  impairment separate
        ▼
   Results / category pages / Alignment / Abatement / Audit
        →  PV, Monte-Carlo P5–P95, tornado, ITR, PCAF, MACC, XLSX + IFRS S2 export
```

3. Layer 1 — Policy & Legal (direct carbon cost)

Module: `carbon_pricing.py`. For each (scenario, region, year, sector):

3.1 Carbon price

`get_carbon_price(scenario, year, band)` linearly interpolates the NGFS trajectory and holds it flat outside 2025–2050, then multiplies by a Monte-Carlo perturbation `price_scale` (1.0 = base):

```
price(y) = interp(NGFS[scenario][band], y) × price_scale
```

Region **band** ∈ {advanced, emerging, rest_of_world} from `get_ngfs_region(ISO3)`. IPCC SSP scenarios map to their closest NGFS analog.

3.2 Abatement pathway (P0)

An entity with a Scope 1+2 net-zero **target_year** decarbonises linearly; carbon cost then applies only to the declining residual:

```
abatement_mult(y) = 1.0                               if no target or y ≤ 2025
                  = residual                           if y ≥ target_year          (residual = re
                  = 1 + (y−2025)/(target−2025)·(residual−1)   otherwise
```

3.3 Cost decomposition

```
direct   = (Scope1 + Scope2) × abatement_mult(y)
pass_through = min(1, sector_PT × pt_scale)      (or per-asset override)
gross    = direct × priced_fraction × price      (priced_fraction = priced_pct/100; free allocat
absorbed = gross × (1 − pass_through)             ← margin compression → firm CF
passed_through = gross × pass_through             ← becomes the L3 market shock
scope3_indirect = Scope3 × price × (1 − pass_through)
net_carbon_opex = absorbed + scope3_indirect      ← what hits the firm's cash flow
```

- **priced_fraction** models free allocation / partial ETS coverage (P0). Default 1.0.
- **Scope-3 mode** (P0): `full` counts scope3_indirect in L1 *and* propagates upstream in L3 (reproduces the methodology's worked examples). `auto` sets the L1 Scope-3 term to 0 whenever L3 is enabled, so upstream cost is counted once.

Data: `carbon_prices_ngfs.json` — real NGFS Phase V REMIND-MAgPIE 3.3-4.8 prices pulled from the IIASA explorer, rebased ×1.18 (US GDP deflator 2010→2020) to ~USD2020. Six scenarios carry real data; `divergent_net_zero` + three IEA scenarios retain Appendix-D placeholders. **Pass-through:** `sector_pass_through.json`, sector medians (Sijm 2012; Fabra & Reguant 2014; Cludius 2020).

4. Layer 2 — Technology (learning curves & stranding)

Module: `learning_curves.py`.

4.1 Wright's Law

```
Cost(Q) = Cost₀ × (Q/Q₀)^b ,    b = log₂(1 − LR)
Q/Q₀ = (1 + g)^(y − 2025) ,     g = max(0, scenario deployment growth)
```

Negative growth (demand decline) is clamped to 0 — cumulative capacity is monotonic; demand decline is handled by the sector pathway, not the cost curve.

4.2 Lafond distributional band

```
σ_t = lafond_σ × √(y − 2025)
cost_lo = pred × e^(−σ_t) ,  cost_hi = pred × e^(+σ_t)
```

4.3 Stranding trigger (either fires)

- **Cost crossover** — first year challenger cost ≤ incumbent cost.
- **Demand collapse** (fossil-dependent sectors only) — first year the sector pathway falls below 0.5.

4.4 Impairment

```
frac(y)  = 1 / (1 + e^(−0.20·(y − crossover)))           logistic, slope 0.20
base     = replacement_value            if fossil-dependent
         = replacement_value × 0.5      otherwise         (only incumbent-tied value can strand)
cap      = max(0, 1 − pathway(2050)) × base               scenario-scaled ceiling
annual_impairment(y) = (frac(y) − frac(y−1)) × cap
```

Impairment is a **balance-sheet** figure, reported separately from cash flow.

4.5 Revenue erosion

```
revenue_index(y) = sector pathway(y)            (production index, 2025 = 1.0)
revenue_erosion(y) = max(0, revenue × (1 − index(y)))     ← cash-flow cost
```

Data (P0 refresh): power-sector base costs are **LCOE (USD/MWh)** from **IRENA Renewable Power Generation Costs in 2024** and **Lazard LCOE+ 2025 v18** (coal 118, gas 76, solar 43, onshore wind 34, storage 170). Learning rates: Way et al. 2022 / IRENA. Industrial base costs remain sector estimates.

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5. Layer 3 — Market (production-network propagation)

Modules: `network_propagation.py` (world), `network_mrio.py` (MRIO + cascade).

5.1 Sectoral shock

Each sector's carbon shock as a fraction of its output:

```
sᵢ = emission_intensityᵢ × price × pass_throughᵢ × 1e−6
```

(emission_intensity in tCO_2 per M$ revenue; 1e−6 → per-USD.) Using sector-typical exposure avoids double-counting the focal asset's own pass-through.

5.2 Leontief propagation

```
L = (I − A)⁻¹
total_shock_j = Σᵢ L[i,j] · sᵢ
propagated_j  = total_shock_j − own       (own = L[j,j]·s_j, already in L1)
indirect_cost = propagated_j × asset_revenue
```

CES damping: off-diagonal A scaled by 1/σ before inversion (σ = substitution elasticity; Papageorgiou 2017 range 1.3–3.0). σ=1 is Cobb-Douglas.

5.3 Resolution (P2)

- **World** (default): 20×20 single-region matrix (EXIOBASE-3 world totals, `io_matrix.json`).
- **Multi-region**: full **20 sectors × 49 regions = 980×980** EXIOBASE MRIO (`io_matrix_mrio.npz`). Cross-border supply chains are explicit and each region's sectors pay carbon on **that region's price band**, so an EU refiner's indirect cost reflects emerging-market upstream at a different carbon price. Leontief inverts in ~0.1 s.

5.4 Endogenous-default cascade (P2, Reisch et al. 2025)

Beyond linear Leontief, nodes absorbing input-cost shock above threshold **θ** pass an amplified shock downstream, iterated to convergence:

```
s_eff = s
repeat:  total = L · s_eff ;  over = max(0, total − θ) ;  s_eff = s + contagion · over
```

Default off; triggers only for the most-exposed nodes under stress (θ≈0.02 → ~1.25× for CHN steel; θ=0.01 → ~6.8×).

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6. Layer 4 — Reputation (CCExposure, z-standardised)

Module: `cc_exposure.py`. **P0 fix (D3):** raw 0–2 proxies inflated elasticities ~500–1000×; the engine now consumes **z-scores** against the real Sautner distribution and prices **per standard deviation**.

```
rawₓ = sector_exposureₓ × scenario_modifierₓ          x ∈ {opp, reg, phy}
zₓ   = (rawₓ − pooled_meanₓ) / pooled_sdₓ
z_total = (Σ rawₓ − pooled_mean_total) / pooled_sd_total

credit_spread_bps = 12·z_reg + 6·z_phy
equity_premium_bps = 50·z_total                        (Pricing paper: premium per 1 SD of overal
revenue_growth_bps = 35·z_opp − 25·z_reg
```

The **pooled distribution** is Sautner et al. (JoF 2023) **Table 1** (firm-year, $\times 10^3$): opp 0.391/1.344, reg 0.049/0.264, phy 0.013/0.103, total 0.943/2.443. **Sector exposures** are anchored to Sautner **Table 4** by SIC industry where available (utilities → power, petroleum refining, transport equipment → autos, primary metal → steel …), estimated from adjacent industries otherwise.

Effect: average-exposure sectors carry ≈0 premium (the inflation is gone); coal keeps ~29 bps credit / −39 bps revenue drag; renewables +85 bps revenue uplift.

Routing (non-duplication): `ROUTE_CASHFLOWS` (default) applies the revenue-growth modifier to cash flows (credit/equity are diagnostic); `ROUTE_WACC` adds credit+equity to the discount rate and zeroes the revenue modifier. **Firm override (P2):** a licensed firm-level CCExposure feed replaces the sector median via `firm_override`.

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7. Orchestration & non-duplication

`run_asset_transition(asset, scenario, …)` composes the layers. Each channel routes once:

```
total_cf(y) = L1 net_carbon_opex
            + L2 revenue_erosion
            + L3 indirect_input_cost
            + L4 revenue_modifier (as a cost: −modifier)
impairment(y) = L2 annual_impairment            (balance-sheet, NOT in total_cf)
wacc_premium_bps = L4 credit+equity              (only if ROUTE_WACC)
```

`run_portfolio_transition` runs all assets × scenarios and returns `{scenario: [TransitionAssetResult]}`. Present value uses `Σ cost / (1 + wacc)^(y − 2025)`.

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8. Uncertainty & sensitivity

8.1 Monte-Carlo (`uncertainty.py`)

Re-runs the four layers over N draws (default 400), perturbing:

```
price_scale        ~ LogNormal(0, 0.20)      carbon-price / policy path
pass_through_scale ~ Normal(1, 0.10)         cost incidence
elasticity σ       ~ Uniform(1.0, 2.5)       network substitution
```

Returns P5 / P50 / P95 of PV transition cost and stranded impairment. Seed-deterministic.

8.2 Tornado (`sensitivity.py`)

One-at-a-time swing of each driver (carbon price ±30%, pass-through ±20%, σ 1.0–2.5, Scope-3 mode, discount ±2pp) with all others at base; sorted by |swing|. Attributes the Monte-Carlo spread to individual assumptions so you know which input to firm up.

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9. Decision & alignment layer

9.1 Financed emissions — PCAF (`alignment.py`)

Attributed emissions = Σ `attribution_pct` × (Scope 1/2/3). 100% = corporate own-asset view; a lender/investor enters their stake.

9.2 Implied Temperature Rise

Each entity's abated Scope 1+2 pathway vs a 1.5 °C linear-to-net-zero budget:

```
budget = 0.5 × E₀ × (2050 − 2025)                        area under a straight line to zero
ratio  = cumulative_actual / budget − 1
ITR    = clamp[1.2, 4.0]( 1.5 + 1.2 × ratio )
portfolio ITR = Scope1+2 × attribution weighted mean
```

Net-zero-by-2050 ≈ 1.5 °C; flat emissions ≈ 2.8 °C.

9.3 Pathway alignment (PACTA-style)

Compares the entity's Scope 1+2 decline by 2050 to the scenario's sector demand pathway; gap ≤ 0.05 → aligned, ≤ 0.25 → lagging, else misaligned.

9.4 Peer benchmarking

Entity economic intensity (tCO_2 Scope 1+2 / $M revenue) vs the **real Sautner firm-level Carbon Intensity distribution** (JoF 2023 Table 1: median 11, p75 84.6 across ~10k firms) → quartile position.

9.5 Marginal abatement cost curves (`macc.py`)

Per-sector measures (cost USD/tCO_2, potential share of Scope 1+2). At carbon price P, all measures with cost ≤ P are cost-effective:

```
abated_frac = Σ potential (cost ≤ P) ,  capped at 1
abatement_cost = Σ cost × potential × emissions
carbon_cost_avoided = abated × P ,  net_benefit = avoided − cost
```

9.6 Budget optimizer (`optimizer.py`)

Merit-order greedy: sort all measures across the portfolio by \$/tCO_2; fund no-regret (negative-cost) measures always, then buy cheapest-first until the budget is exhausted. Returns abated tonnes, net spend, marginal-cost frontier, residual carbon cost, and per-asset allocation. Optimal for max tonnes per dollar under a linear MACC.

9.7 Disclosure export

`build_disclosure_report` generates a Governance → Strategy → Risk Management → Metrics report mapped to **IFRS S2** clauses and **ESRS E1** datapoints, plus a provenance-stamped multi-sheet XLSX.

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10. Data provenance & vintages

Component	Source	Vintage
Carbon prices (L1)	NGFS Phase V REMIND-MAgPIE 3.3-4.8 (IIASA)	Nov 2023, ×1.18 → USD2020
Pass-through (L1)	Sijm 2012; Fabra & Reguant 2014; Cludius 2020	sector medians
Learning rates (L2)	Way et al. 2022; IRENA	2022–23
LCOE base costs (L2)	IRENA RPGC 2024; Lazard LCOE+ 2025 v18	2024–25
I-O matrix (L3)	EXIOBASE-3 (Stadler et al. 2018), IOT 2019	2019
CCExposure + z-base (L4)	Sautner et al. JoF 2023 Tables 1 & 4	2002–2019, 10k firms
Sector pathways	NGFS / IEA WEO 2023	2023
MACC	IPCC AR6 WG3; IEA roadmaps	2022–23

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11. Calibration — worked examples (default settings, Scope-3 full)

- **Coal plant** (\$500M, 2.5 MtCO_2): Net-Zero-2050 L1-2050 ≈ **\$169M**; cumulative impairment **\$266.1M** / 97.3% stranded (crossover 2025). Current Policies ≈ \$7M / \$33M.
- **Oil refinery** (\$2B, 12.8 MtCO_2): stranding triggered by **demand collapse** (crossover 2039), not cost crossover.
- **Office** (\$50M, 1k tCO_2): no stranding; small net cost / reputational uplift.

These are unchanged by the P0–P3 upgrades (abatement/priced default to legacy; world L3 default; LCOE preserves the 2025 crossover).

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12. Known limitations (challenge these)

- **Pass-through** is a static sector median; firm-specific values vary with market power and trade exposure. Override per asset.
- **CCExposure** sector exposures are anchored to published SIC-industry means, estimated for sectors not in Table 4; the licensed firm-level feed is the fix.
- **MRIO twin sectors** (`road_transport_ev`, `real_estate_residential`) are supply-split from their siblings (EXIOBASE has no separate EV / residential-RE industry).
- **Cascade θ / contagion** are judgment parameters; the linear result is the default.
- **ITR / MACC / peer benchmark** are screening approximations aligned to SBTi/PACTA/PCAF/AR6 in spirit, not certified implementations.
- **Carbon-price vintage** is NGFS Phase V (Nov 2023); refresh for disclosure.
- Emissions decarbonise only if a target is set; without one, Scope 1+2 are held flat.

Appendix A — Sector taxonomy (20 sectors)

Key · label · incumbent → challenger · fossil-dependent · emission intensity (tCO_2/M$). See `data/transition/sector_taxonomy.json`. 20 sectors span power (coal/gas/renewable), oil & gas (upstream/refining/distribution), heavy industry (steel/cement/chemicals/aluminium), transport (road ICE/EV, aviation, shipping), real estate (commercial/residential), agriculture, data centres, general manufacturing, and services.

Appendix B — Pass-through coefficients

See `data/transition/sector_pass_through.json` (pass-through, demand elasticity, market structure, trade exposure, source).

Appendix C — Learning rates & LCOE

See `data/transition/learning_curves.json`. Power LCOE (USD/MWh): coal 118, gas 76, solar 43, onshore wind 34, storage 170 (IRENA 2024 / Lazard 2025).

Appendix D — NGFS carbon prices

See `data/transition/carbon_prices_ngfs.json`. Real REMIND-MAgPIE, three bands, 2025–2050.

Appendix E — Layer-4 z-base & elasticities

Pooled distribution (Sautner Table 1, $\times 10^3$): opp 0.391/1.344 · reg 0.049/0.264 · phy 0.013/0.103 · total 0.943/2.443. Per-SD elasticities (bps): equity 50·z_total; credit 12·z_reg + 6·z_phy; revenue 35·z_opp − 25·z_reg.

Appendix F — MACC measures

See `data/transition/macc.json`. Per-sector measures with marginal cost (USD/tCO_2) and abatement potential (share of Scope 1+2); AR6 WG3 / IEA-informed.

Appendix G — References

NGFS Phase V (IIASA) · Sijm 2012 · Fabra & Reguant 2014 · Cludius 2020 · Way et al. Joule 2022 · Lafond et al. TFSC 2018 · IRENA RPGC 2024 · Lazard LCOE+ 2025 · Stadler et al. (EXIOBASE) 2018 · Acemoglu et al. 2012 · Papageorgiou et al. 2017 · Reisch et al. 2025 (arXiv:2503.10644) · Sautner, van Lent, Vilkov, Zhang (JoF 2023; Mgmt Sci 2023) · IPCC AR6 WG3 · PCAF 2022 · SBTi · PACTA.